

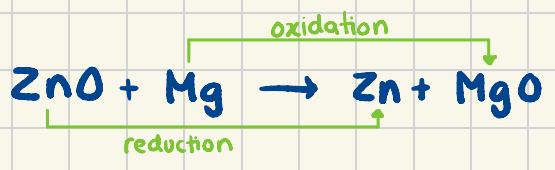
6 - Electrochemistry

15/12/24

Oxidation & Reduction

- oxidation is the gain of oxygen by an element
- reduction is the loss of oxygen by an element
- * oxidation & reduction always take place together

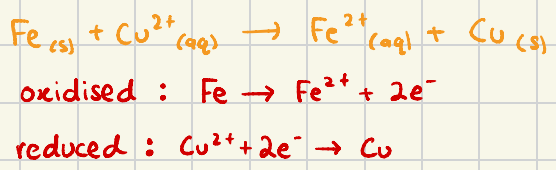
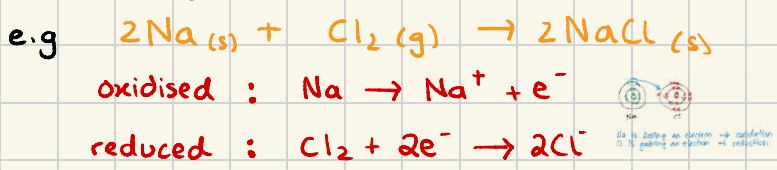
Oxidation	Loss of e ⁻	Reduction	Gain of e ⁻
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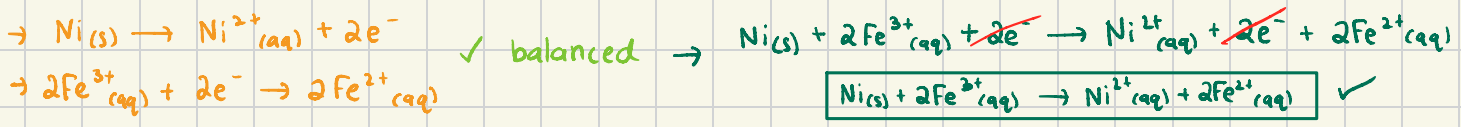
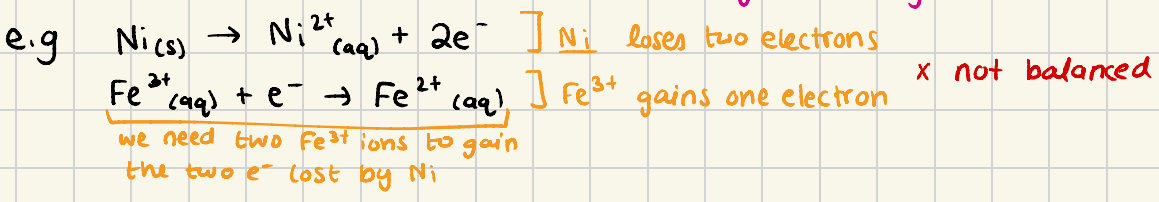
REDOX REACTION

Half-equations

→ Separating a reaction into two separate reactions, one showing oxidation & the other reduction



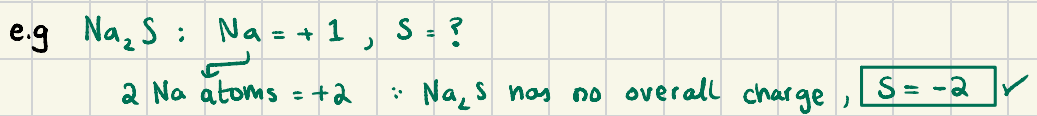
⇒ we can make balanced ionic equations by balancing the half-equations



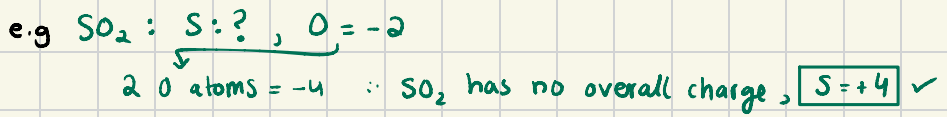
Oxidation number [abbreviation : ox. no.]

- a number given to each atom/ion in a compound that shows us its degree of oxidation
- can be +, -, or 0
- higher +ve number means more oxidation, higher -ve number means more reduction

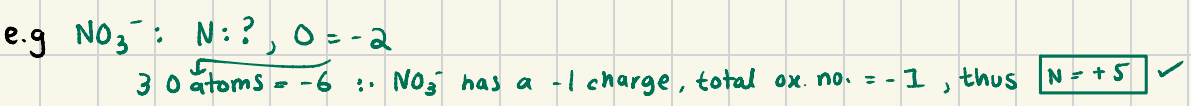
* a metal with a non-metal : the metal has the +ve ox. no. & the non-metal has the -ve ox. no.



* a non-metal with a non-metal : the more electronegative element is given the -ve ox. no.



* compound ions : use rule number 5 to work out the ox. no. we don't know



⇒ oxidation number rules

- 1) oxidation number of an uncombined element is zero
- 2) in compounds, many atoms/ions have fixed oxidation number:
 - ↳ Group 1 elements are always +1
 - ↳ Group 2 elements are always +2
 - ↳ Fluorine is always -1
 - ↳ hydrogen is always +1 (except in metal hydrides, where its -1)
 - ↳ oxygen is always -2 (except in peroxides, where its -1 & in F_2O , where its +2)
- 3) oxidation number of a monatomic ion is the same as its charge
- 4) sum of oxidation number in a compound is zero
- 5) sum of oxidation number in an ion is equal to its charge
- 6) the more electronegative element is given the negative oxidation number

⇒ oxidation numbers are written as roman numerals

e.g. $CrO \rightarrow$ chromium (II) oxide, $Cr_2O_3 \rightarrow$ chromium (III) oxide, $CrO_3 \rightarrow$ chromium (VI) oxide

Oxidising & reducing agents

→ oxidising agent: oxidises another substance but itself is reduced

- increases the oxidation number of another atom/ion
- when that happens, its own oxidation number decreases

e.g. oxygen, chlorine, potassium manganate (VII)

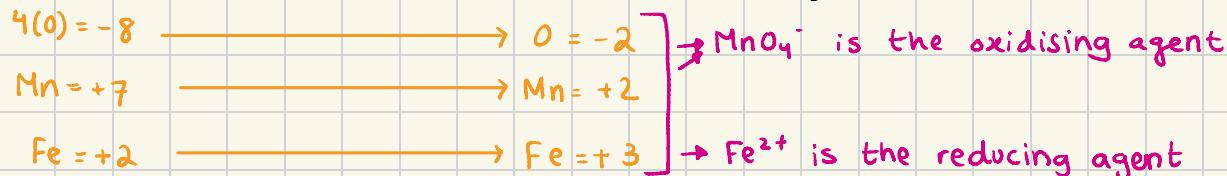
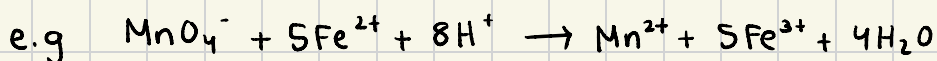
→ reducing agent: reduces another substance but itself is oxidised

- decreases the oxidation number of another atom/ion
- when that happens, its own oxidation number increases

e.g. hydrogen, potassium iodide & reactive metals (such as aluminium)

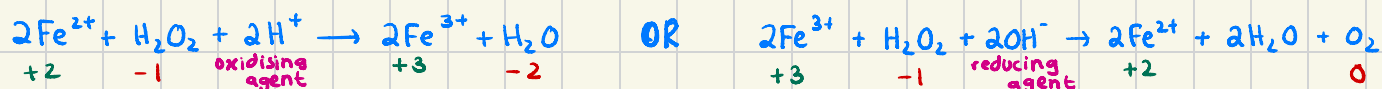
* oxidation is an increase in oxidation number

* reduction is a decrease in oxidation number



* many substances can act as either oxidising or reducing agents depending on the substances they react with as well as the conditions

e.g. hydrogen peroxide

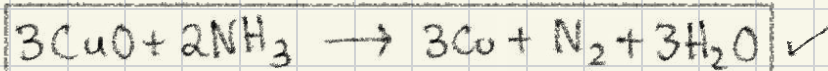
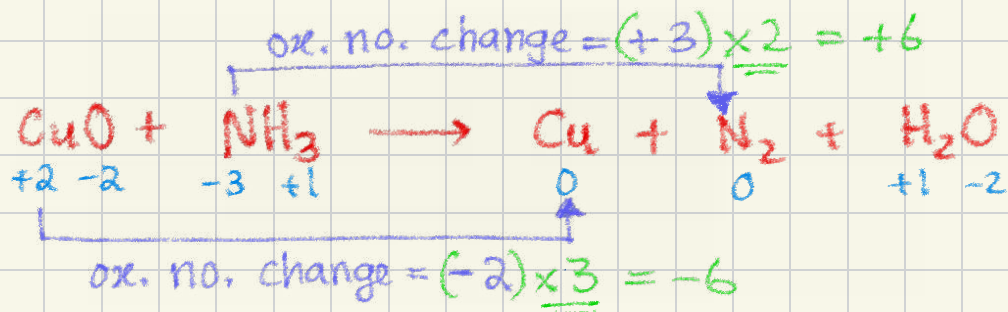


Constructing equations using oxidation numbers:

- 1) identify oxidation number changes
- 2) balance oxidation number changes
- 3) balance charges
- 4) balance atoms

Copper (II) oxide $[CuO]$ reacts with ammonia $[NH_3]$ to form copper, nitrogen $[N_2]$ and water.

Deduce the balanced equation:



Disproportionation reaction

* a reaction in which the same element is simultaneously oxidised & reduced

e.g

